

The Living World....

SHORT NOTES

Features of Living organism

- Growth
- Reproduction
- Metabolism
- Self consciousness

Note: Growth, Metabolism and self consciousness is defining property of living organism.

* Growth → Increase in mass
→ Increase in number

* Reproduction - Some living organism unable to reproduce like mules, sterile worker bees etc. that's why it is not defining property of living organism.

* Metabolism and cellular organisation - Metabolic reaction occur inside the cell, it means cellular organization is strictly required for metabolism to define the feature.

* Consciousness - Ability to sense their environment.

Taxonomy

- Identification - Correct description of orgm.
- Nomenclature - Scientific Naming
- Classification - Give a particular position of an organism in a particular ~~area~~ taxa.

ICBN: International Code for Botanical Nomenclature

ICZN: International code for Zoological Nomenclature

* Binomial Nomenclature was given by Carolus Linnaeus, It contain name with two components: Generic Name and Specific Epithet.

Ex: Mangifera indica

Generic name specific epithet

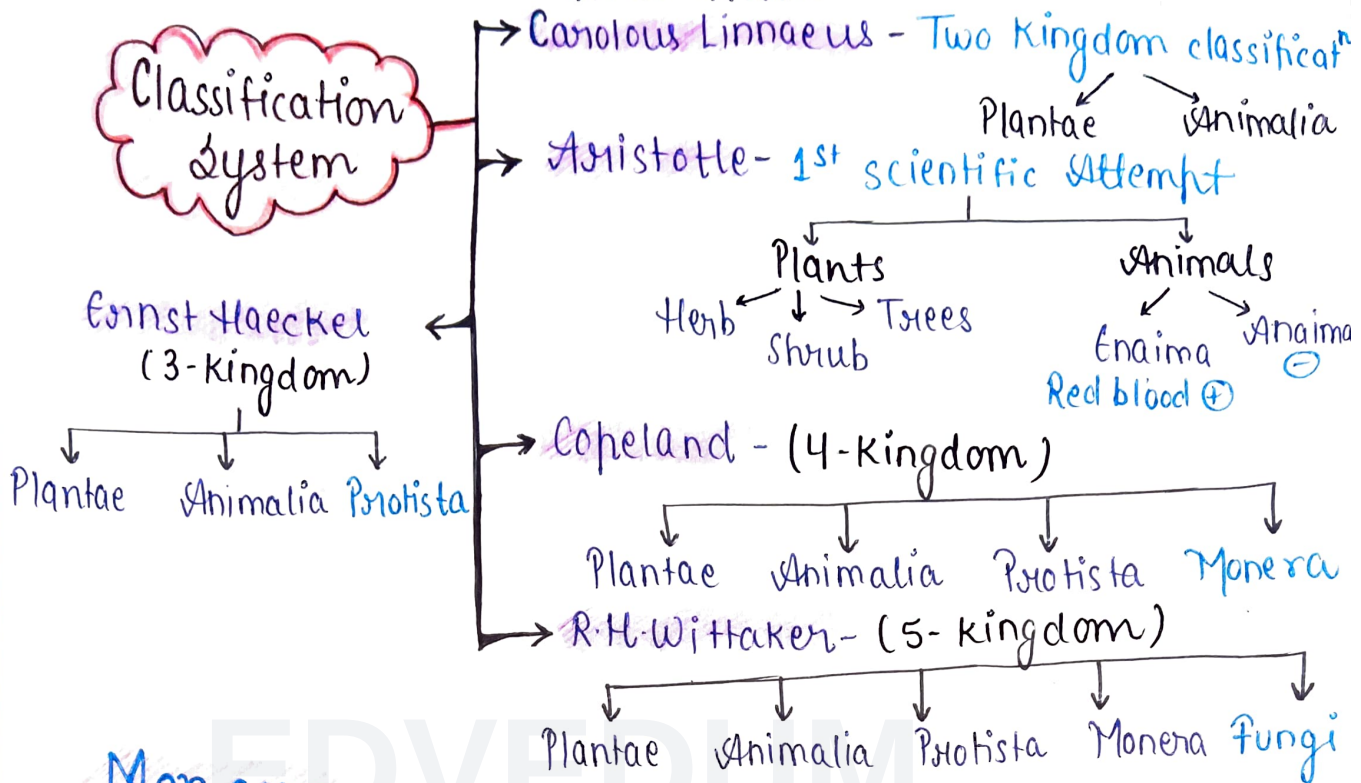
Taxonomic Categories....

Kingdom → Phylum → Class → Order → Family → Genus → Species
(Keep) (Podium) (Clean) (nisi) (Fam) (Get) (sick)

Trick:

Biological Classification...

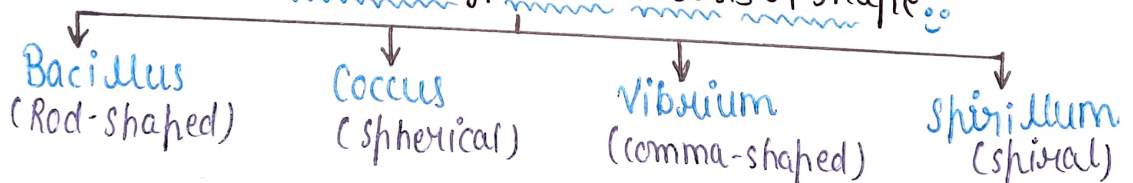
SHORT Notes



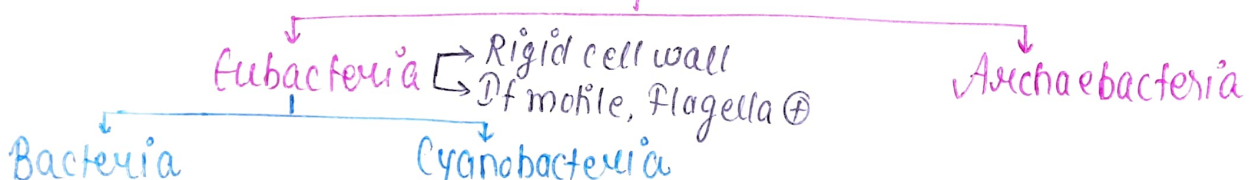
Monera...

- All are Prokaryotes
- Bacteria are the sole members
- Found almost everywhere
- Also present in extreme habitats like desert, snow, hot spring etc.
- Bacteria str. simple but complex in behaviour
- Most extensive metabolic diversity

Bacteria - 4 Types on the Basis of Shape



Kingdom Monera

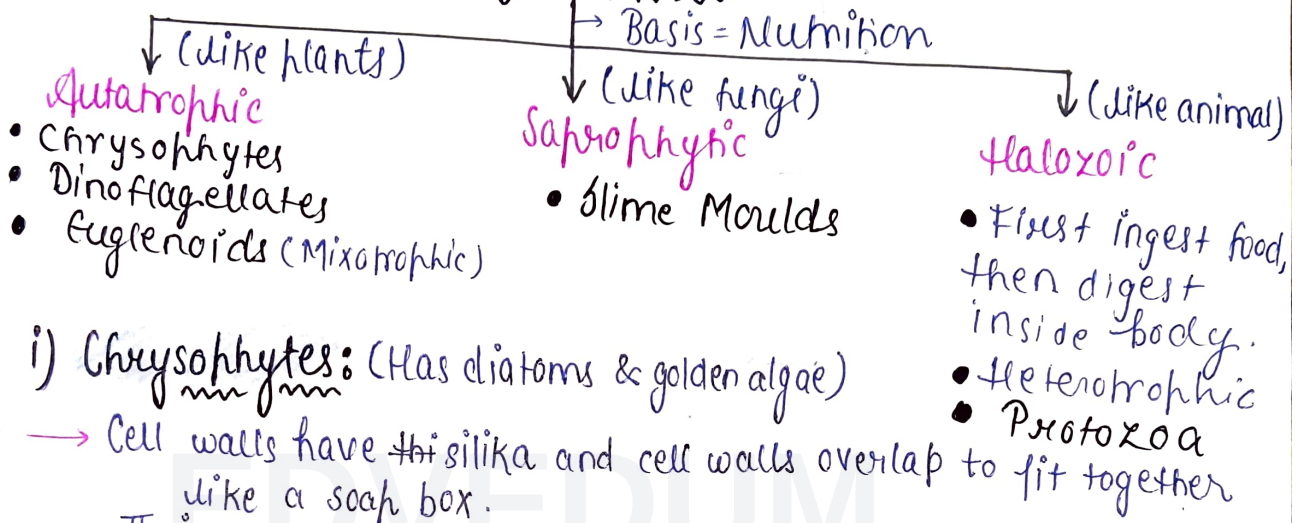


Mycoplasma: Completely lack cell wall smallest living cells, can survive without oxygen. Pathogenic in animals and plants.

Protista...

- All single celled eukaryotes.
- Forms a link b/w plants, animals & Fungi.

Kingdom Protista



i) Chrysophytes: (Has diatoms & golden algae)

- Cell walls have ~~thi~~ silica and cell walls overlap to fit together like a soap box.
- Their accumulation form 'Diatomaceous Earth'
- Used in polishing, filtration of oil and syrups.

ii) Dinoflagellates:

- Marine, photosynthetic, cell wall has cellulose. Two flagella one longitudinal and other transversely in a furrow b/w wall plates.

iii) Euglenoids:

- Have protein rich layer "pellicle" which makes body flexible. Photosynthetic in presence of sunlight but become heterotrophs if they do not get sunlight.

iv) Slime Moulds: Saprophytic protist form aggregate to form plasmodium grows on decaying twigs and leaves. Spores have true walls which are extremely resistant and survive for many years.

- V) Protozoans
- Amoeboid - Amoeba
 - Flagellated - Trypanosoma
 - Ciliated - Paramecium
 - Sporozoans - Plasmodium

Fungi...

- Non-chlorophyllous hyphae
- Network of hyphae called mycelium.
- Cell wall of chitin and polysaccharide
- Grow in warm and humid places.

i) Phycomycetes

- * Grow on decaying wood
- * Mycelium Aseptate
- * Endogenously
- Ex: Rhizopus
Albugo



ii) Ascomycetes

- * Sac fungi
- * Mycelium-branched Aseptate
- * Asexual spores called 'conidia' produced by exogenously
- * Ex: Aspergillus
Neurospora

iii) Basidiomycetes

- * Mycelium-septate
- * Exogenously produced on basidium
- * Ex: Agaricus
Ustilago
Puccinia

iv) Deuteromycetes

- * Fungi imperfect as sexual form
- * Produced asexual spores called 'conidia'
- * Ex: Alternaria
Colletotrichum
Trichoderma

Viruses...

- Inert outside host cell.
- Obligate parasites
- Nat 'nearly living'
- Non-cellular.

* Animal virus - ssRNA, dsRNA, dsDNA

* Plant virus - ssRNA

* Bacterial virus - dsDNA

→ W. M Stanley - crystallised virus
Crystal of protein

→ Beijerinck - Contagium vivum fluidum
Term - 'Virus'

→ Dmitri Ivanovsky - Cause Mosaic disease tobacco.

Viroids...

- * Smaller than Virus
- * Free RNA (low molecular weight)
- * Lack protein coat
- * Cause: Potato Tuber Spindle dis.

Prions...

- * Abnormally folded protein
- * No genetic material
- * Similar size as Virus
- * Mad cow disease or BSE
- * Human: CJD disease

Plant Kingdom...

SHORT NOTES...

Natural Classification

- * By George Bentham and Hooker.
- * Based on Natural affinities among org.

Phylogenetic Classification

- * Based on evolutionary relationship b/w the various organism.
- * Assumes that orgm. belonging to the same taxa have a common ancestor.

Classification System...

Artificial System

- * Given by Carolus Linnaeus
- * Based on - vegetative & Androecium structure
- * Separated by closely related species since they were based on few characteristic -
- * Equal weightage to vegetative and sexual characteristic

Algae...

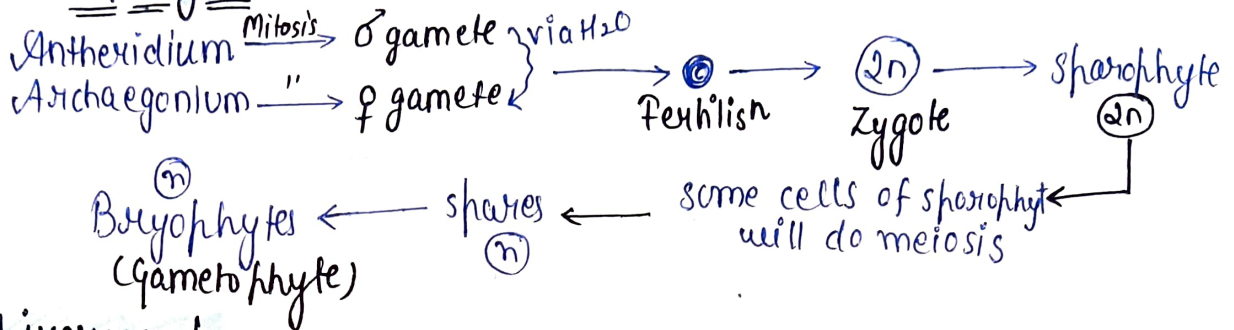
Features	Classification of Algae		
	Chlorophyceae	Phaeophyceae	Rhodophyceae
Common Name	Green Algae	Brown Algae	Red Algae
Pigments	Chl a, b	Chl a, c & fucoxanthin	Chl a, d and γ -phycoerythrin
Reserve food	Starch	Laminarin, Mannitol	Floridean starch
Flagella	2-8, apical, equal	2, unequal lateral	Absent
Examples	Chlamydomonas, Volvox	Ectocarpus, Sargassum	Polysiphonia, Gracilaria

Bryophytes...

- Commonly called as Amphibian of Plant Kingdom.
- Includes Liverworts and mosses
- Vascular Tissue are Absent
- Lack true stem, leaves and roots.
- Main plant body is Gametophyte. Sporophytes is dependent on gametophyte for nutrition.
- Water is essential for fertilisation.

→ Plant body attached to substratum by unicellular or multicellular rhizoids.

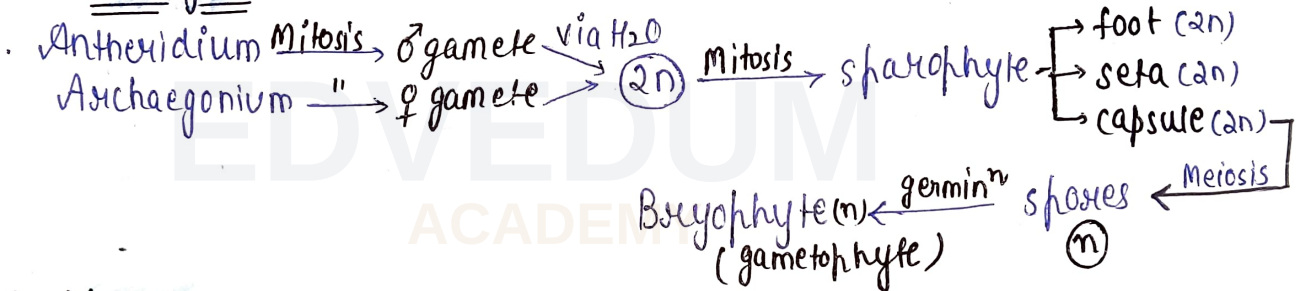
Life cycle:



i) Liverworts:

- * Grow usually in moist and shady habitats; ex: *Marchantia*
- * Plant body is thalloid which is dorsiventral and closely appressed to the substrate.
- * Leafy member have tiny leaf-like appendages in two rows on the stem-like structure.
- * **Gametes** are green, multicellular asexual buds.

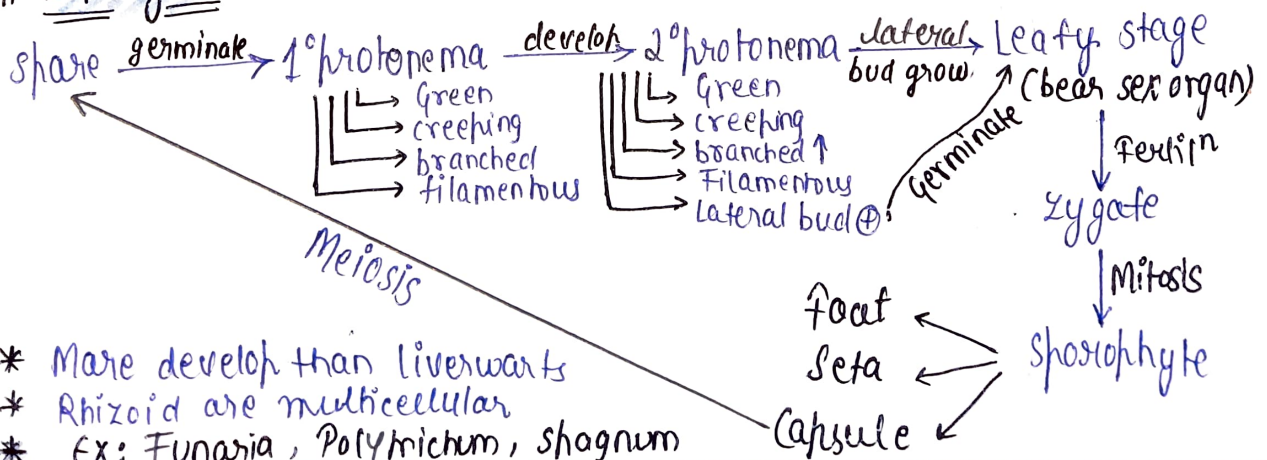
Life cycle:



ii) Mosses:

- * Main plant body of mosses is sporophyte (n)
- * Life cycle of mosses - First stage - Protonema
second stage - Leafy stage

Life cycle:



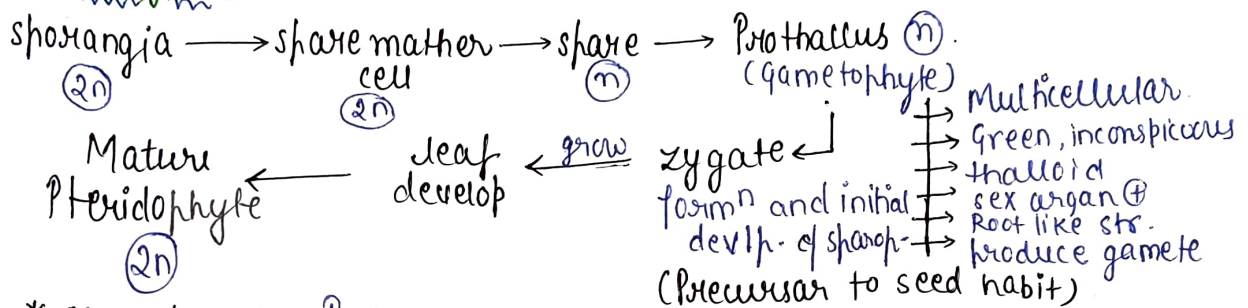
- * More develop than liverworts
- * Rhizoid are multicellular
- * Ex: *Funaria*, *Polytrichum*, *Shagnum*

Pteridophytes.....

- * Includes horsetails and ferns.
- * First vascular plants without seeds.
- * Main plant body is sporophyte and distinguish into root, stem and leaves.
- * Male gametes are flagellated.
- * Requires water for fertilisation.

Note: Selaginella & Equisetum form cone/strobilus.

Life cycle:



- * Macrophyte: Ferns
- * Microphyte: Selaginella
- * Heterosporous - Selaginella, Salvinia

Classification of Pteridophytes:

- Psilopsida - Psilohum
- Pteropsida - Pteris, Dryopteris, Adiantum
- Lycopsidea - Lycopodium, Selaginella
- Sphenopsida - Equisetum

Gymnosperm.....

- * Naked seed
- * No ovary, No fruit
- * Non-thalloid
- * Heterosporous

Ex: Cycus

- Unbranched
- Unisexual
- Pinnate leaves
- Tolerate Temp↑

Pinus

- Branched
- Bisexual
- Needle like leaves
- Sunken stomata
- Cuticle↑

* Tallest tree: Sequoia Gigantica

* Association: Mycorrhiza with Pinus
Caralloid coat: Cycus, cyanobacteria

Angiosperm.....

- * Angiosperm are flowering plant and have seeds enclosed in fruits and well developed flowers containing reproductive organ.
- * Smallest tree: Wolffia
- * Tallest tree: Eucalyptus
- * Types ↗ Monocotyledon
↘ Dicotyledon

Animal Kingdom

SHORT Notes

* Types of Symmetry ::

- Asymmetrical: Sponges
- Radially Symmetrical: Coelenterata, Ctenophora & Adult Echinoderms
- Bilaterally Symmetrical: Annelids, Arthropods, Platyhelminthes, Larva of Echinoderms, Hemichordates and Chordates.

* Level of Organisation ::

- Cellular organisation: Sponges
- Tissue level: Coelenterata and ctenophora
- Organ Level: Platyhelminthes
- Organ System level: Annelida to chordates.

* Complete Digestive System - Starts from Aschelminthes to Chordates

* Closed Type of Circulatory System - Annelids, Cephalopod, Molluscs and vertebrates.

Coelom ::

- Acoelomates - Parifera to Platyhelminthes
- Pseudo coelomates - Aschelminthes
- Eucoelomates - Annelids to chordates

* Segmentation: Annelids, Arthropods and Chordates

Special features

- 1) Porifera - Water transport or Canal system
 - 2) Echinodermata - Water vascular system or Ambulacral system
 - 3) Coelenterata - Metagenesis or Alternation of generation - Ex: Obelia
 - 4) Ctenophores - Bioluminescence and all organism are marine.
 - 5) Platyhelminthes - Hooks and suckers are present in parasitic forms.
 - 6) Aschelminthes - Muscular pharynx, male (smaller + curved)
Female (Longer + straight)
- * Parapodia (locomotory organ) of aquatic annelids like Nereis which help in swimming.
- * Arthropoda - Largest phylum
- * Mollusca - 2nd Largest phylum
- Dioecious
 → Clam worm

- * Arthropoda - Chitinous exoskeleton & Blood in colourless (Haemolymph)
- * Mollusca - Presence of (Ctenidia + organ of Bojanus + heart form)
- * Central Nervous System:
 - Non-chordates - Solid + Double + Ventral side of body
 - Chordates - Hollow + Single + Dorsal side of body
- * Urochordata: Retrogressive Metamorphosis
- * Cephalochordates - Progressive Metamorphosis
- * Heart Chambers - Pisces-2, Amphibians-3, Reptiles-3, (Crocodiles-4 exception) Aves-4, Mammals-4
- * Chondrichthyes - Cartilaginous Fishes and marine
- * Osteichthyes - Bony fishes & Both marine and fresh water fishes.

Reptiles

- Exoskeleton is made up of bony epidermal scales or scutes
- Skin is dry, cornified, rough & non-glandular.

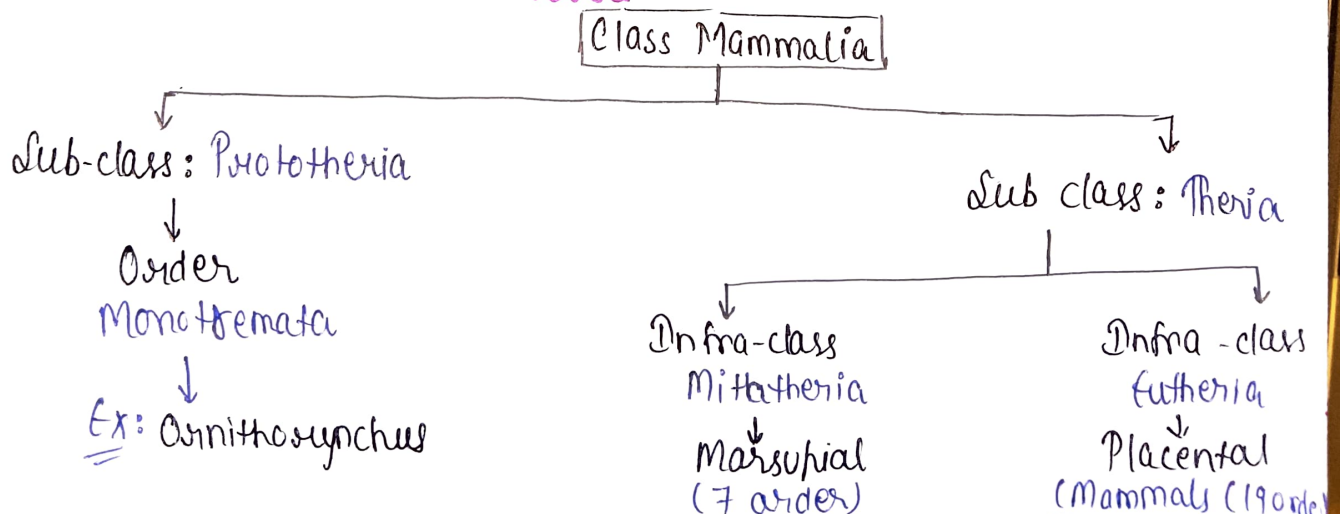
Aves

- Feather present
- Forelimbs modified into wings.
- Hindlimbs have scales and modified for walking swimming or clamping
- Pneumatic bones

Mammals

- Present of hair & mammary glands
- External ears or pinnae are present
- Homiothermic - Aves, Mammals

Classification of Mammals



Structure of Atom...

SHORT NOTES...

Electromagnetic Spectrum...

- * Radio wave $\rightarrow$ Microwave $\rightarrow$ Infrared rays $\rightarrow$ Visible rays $\rightarrow$ Ultraviolet rays $\rightarrow$ X-rays $\rightarrow$ Cosmic rays
- * Wavelength decreases $\rightarrow$
- * Frequency Increases $\rightarrow$
- * $c = \nu \lambda$ $\rightarrow$ wavelength
Speed of light $\rightarrow$ Frequency
- $\lambda = \frac{c}{\nu}$
- $\nu = \frac{1}{\lambda} = \frac{v}{c}$ $\rightarrow$ wave number
- $T = \frac{1}{\nu}$ $\rightarrow$ Time period
- $E = \frac{hc}{\lambda} = h\nu$ $\rightarrow$ Energy
 $h = 6.626 \times 10^{-34} \text{ Js}$
- $E(\text{ev}) = \frac{12400}{\lambda(\text{\AA})}$

- * Total amount of energy transmitted $E = nh\nu = \frac{nhc}{\lambda}$

Bohr's Atomic Model...

$\rightarrow$ Based on quantum theory of radiation.

- * Radius - $r = 0.529 \times \frac{n^2}{Z} \text{\AA}$
- * Velocity - $v = 2.188 \times 10^6 \frac{Z}{n} \text{ ms}^{-1}$
- * Total Energy = Energy - $-13.6 \times \frac{Z^2}{n^2} \text{ eV/atom}$
- * T.E = $-\frac{KZe^2}{2a}$, P.E = $-\frac{KZe^2}{a}$, K.E = $\frac{KZe^2}{2a}$
- * P.E = $-2KE$, $KE = -TE$, $PE = 2TE$
- * Revolution per second = $\frac{v}{2\pi r} \propto \frac{Z^2}{n^3}$
- * Time for 1 revolution = $\frac{2\pi r}{v} \propto \frac{n^3}{Z^2}$
- * Energy difference b/w n_1 & n_2 energy level is given by -
 $\Delta E = E_{n_2} - E_{n_1} = 13.6 Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ eV/atom}$

$$\begin{aligned} 1 \text{ eV} &= 3.83 \times 10^{-23} \text{ Kcal} \\ 1 \text{ eV} &= 1.602 \times 10^{-12} \text{ erg} \\ 1 \text{ eV} &= 1.602 \times 10^{-19} \text{ J} \end{aligned}$$

$$E = -313.6 \frac{Z^2}{n^2} \text{ Kcal/mole}$$

(1 cal = 4.18 J)

$$\begin{aligned} v &\propto \frac{Z}{n} \\ r &\propto \frac{n^2}{Z} \\ E &\propto \frac{Z^2}{n^2} \end{aligned}$$

$\rightarrow$ M. Imp

Hydrogen Spectrum....

* Rydberg's Equation: $\frac{1}{\lambda} = \bar{\nu} = R_H \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right] \times Z^2$

R_H = Rydberg constant = 109700 cm^{-1}

* For 1st line of series, $n_2 = (n_1 + 1)$

* Limiting spectral line (series limit) means $n_2 = \infty$

* H α line means $n_2 = n_1 + 1$; also known as line of longest λ , shortest $\bar{\nu}$, least E

* Similarly, H β line means: $n_2 = n_1 + 2$

* No. of spectral lines = $\frac{n(n-1)}{2}$
($n \rightarrow$ G-state)

* No. of spectral lines = $\frac{(n_2 - n_1)(n_2 - n_1 + 1)}{2}$
($n_2 \rightarrow n_1$)

* No. of spectral series in a particular series = $n_2 - n_1$

De-Broglie Hypothesis....

* All material particles possess wave character as well as particle character.

$$\lambda = \frac{h}{mv} = \frac{h}{p} \rightarrow \begin{array}{l} \text{Planck constant} \\ \text{momentum} \end{array}$$

* The circumference of n^{th} orbit = $2\pi r_n = n\lambda$
↳ no. of waves

Heisenberg Uncertainty Principle...

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi} \quad \text{or} \quad \Delta x \cdot \Delta v \geq \frac{h}{4\pi m}$$

Δx = Uncertainty in position
 Δp = " " momentum
 Δv = " " velocity
 m = mass of particle.

* Heisenberg replaced the concept of orbit by that of orbital.

For Hydrogen like atom -

$$1s < 2s = 2p < 3s = 3p = 3d < 4s = 4p = 4d = 4f \dots$$

For Multi e^- atom -

$$1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p < 5s \dots$$

Nodes - Probability of finding $e^- = 0$

↳ Radial / spherical nodes = $n - l - 1$
↳ Angular nodes = l
↳ Total nodes = $n - 1$

→ Principle Quantum Number (n)

* Tells About orbit

→ Azimuthal Quantum Number (l)

* Tells about shape and no. of subshell

* Value of (l) for a given $n = 0$ to $(n-1)$

Subshell

s-subshell $\longrightarrow 0$

p-subshell $\longrightarrow 1$

d-subshell $\longrightarrow 2$

f-subshell $\longrightarrow 3$

→ Angular momentum of a subshell = $\sqrt{l(l+1)} \frac{h}{2\pi}$

→ Magnetic Quantum No.

* Tells about orientation of orbital in subshell

* Value of $m = (-l$ to $+l)$

→ Spin Quantum Number

$\frac{1}{2}$

$s = (+\frac{1}{2})$

$s = (-\frac{1}{2})$

→ Magnetic Moment -

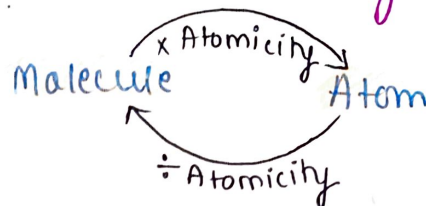
$$\sqrt{\frac{n(n+2)}{4}} \text{ BM}$$

unpaired e^-

Some Basic Concept of Chemistry...

SHORT NOTES.....

* Atomicity: No. of atom in a molecule



$$m_n > m_p >>> m_e$$

$$m_n \approx m_p >> m_e$$

* Mass of an Atom = $m_{\text{proton}} \times (\text{mass no.})$

* Actual mass of an atom = $\text{Mass no.} \times (1.67 \times 10^{-27} \text{ kg})$
 $= \text{mass no.} \times (1.67 \times 10^{-24} \text{ g})$
 $= (\text{mass no.}) \times \text{amu.}$

$$1 \text{ amu} = 1.67 \times 10^{-24} \text{ g}$$

4mp # $N_A \times 1 \text{ amu} = 1$
 (ing)
 $\downarrow$ Avagadro no.

* 1 mole = $6.022 \times 10^{23} \approx 6 \times 10^{23} = \text{NA}$

* 1 mole atom = $1 \times 6.022 \times 10^{23} \text{ atom} = 6 \times 10^{23} \text{ atom} = \text{NA atom}$

* 7 mole molecule = $7 \times 6.022 \times 10^{23} \text{ molecule} = 7 \text{ NA molecule}$

Ex: $\left(\begin{smallmatrix} 16 \\ \text{O} \\ 8 \end{smallmatrix} \right) \rightarrow 16 \text{ amu} = 16 \text{ u} = \text{mass of 1 atom of O}$
 $\rightarrow 16 \text{ g} = \text{mass of 1 mole of atom of O}$

$$1 \text{ u} = 1.66 \times 10^{-24} \text{ g} = \frac{1}{N_A} \text{ g}$$

$\rightarrow$ same no. of proton

* Avg. atomic mass = $\frac{\sum \% \text{ abundance} \times \text{Isotopic mass}}{100}$

* No. of moles (n) = $\frac{\text{Given mass}}{\text{Molar mass}} \times 100$; No. of moles (n) = $\frac{\text{Given no. of particle}}{N_A}$
 $\hookrightarrow$ valid for all $\hookrightarrow$ for molecule = $\frac{n}{N_A}$

Que Calculate no. of moles of 'C' atom present in 88g of CO_2 ?

Soln: $n_{\text{CO}_2} = \frac{88}{44} \text{ g} = 2 \text{ mole of } \text{CO}_2 = 2 \text{ NA molecule of } \text{CO}_2$

Atom of C = $1 \times 2 \text{ NA} = 2 \text{ NA} = 2 \text{ mole}$

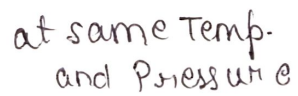
Atom of O = $2 \times 2 \text{ NA} = 4 \text{ NA} = 4 \text{ mole}$

Que Find No. of atom present in 0.016g of methane (CH_4)?

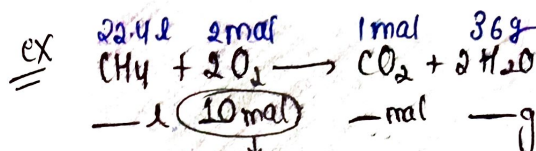
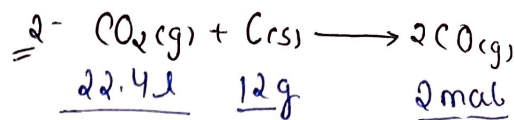
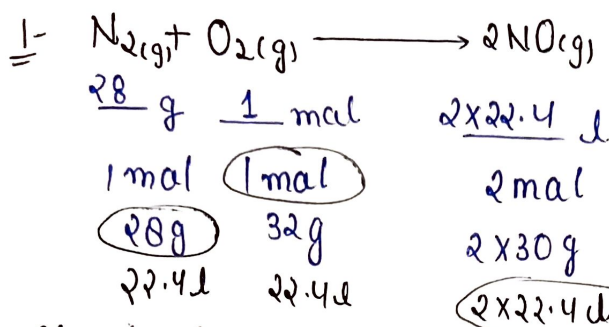
$$\text{No. of atom} = \frac{0.016}{16} \times N_A \times 5 = 0.005 \text{ NA An}$$

Molar Volume of a gas at constant Temp and Pressure is independent of-

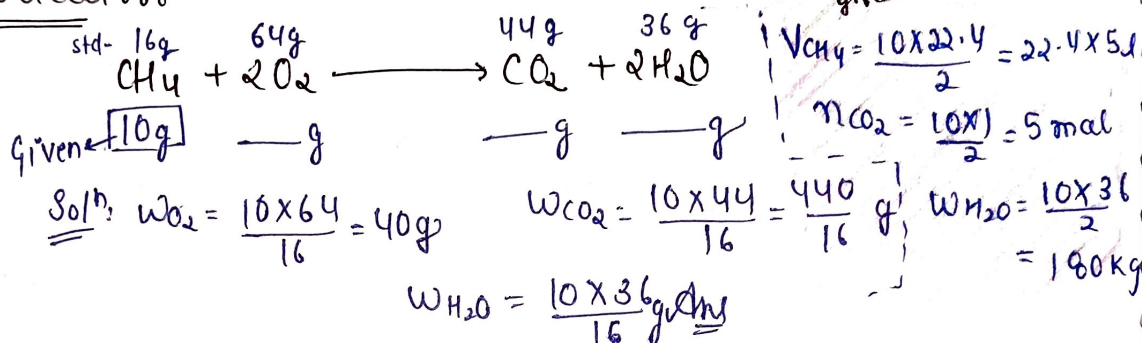
- 1) Nature of gas
- 2) Atomicity of gas
- 3) Molar mass of gas.



Stoichiometric coefficient...



Shortcut...



$$V_{\text{CH}_4} = \frac{10 \times 22.4}{2} = 22.4 \times 5 \text{ l}$$

$$n_{\text{CO}_2} = \frac{10 \times 1}{2} = 5 \text{ mol}$$

$$w_{\text{H}_2\text{O}} = \frac{10 \times 36}{2} = 180 \text{ g}$$

Purity Concept...

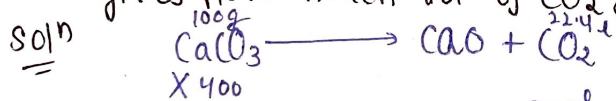
* $\boxed{\text{Mass of given sample of reactant} \neq \text{mass of pure reactant}}$

* $\boxed{\text{Mass of given sample of reactant} = \text{mass of pure reactant} + \text{mass of impurities}}$

* $\% \text{ purity} = \frac{\text{Mass of pure reactant}}{\text{Mass of given sample}} \times 100$

* $\left\{ \text{Mass of pure reactant} = \text{mass of given sample} \times \frac{\% \text{ purity}}{100} \right\}$

Que A sample of 400 g CaCO_3 (80% purity) on decomposition gives how much vol. of CO_2 ?



$$\text{Pure CaCO}_3 = \frac{400 \times 80}{100} = 320 \text{ g}$$

$$\begin{aligned} V_{\text{CO}_2} &= \frac{320 \times 22.4}{100} \\ &= 3.2 \times 22.4 \text{ l} \end{aligned}$$

Q=2 A sample of 200g CaCO_3 sample gives 22.4L of CO_2 on decomposition. what will be the % purity of sample?



pure $\text{CaCO}_3 = 100\text{g}$

$$\% \text{ purity} = \frac{100}{200} \times 100 = 50\% \text{ Ans}$$

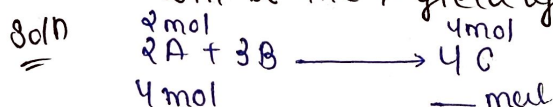
Yield Concept....

Theoretical yield > Actual yield

→ Theoretical yield: calculated from stoichiometry
→ Actual yield: observed yield

$$\% \text{ yield or efficiency} = \frac{\text{Actual yield}}{\text{Theoretical yield}} \times 100$$

Que $2\text{A} + 3\text{B} \longrightarrow 4\text{C}$; if 4 moles of A produces 6 moles of C. what will be the % yield of rean?



Actual yield = 6 mol of C

$$\text{Theoretical yield} = \frac{4 \times 4}{2} = 8 \text{ mole}$$

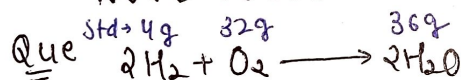
$$\% \text{ yield} = \frac{6}{8} \times 100 = 75\% \text{ Ans}$$

Limiting Reagent....

Identify - reagent, calculate (given no. of mole of reagent / their stoichiometric coefficient)

- * Concept of L-R applied when quantity of more than one reactant are given
- * Control product formation.

Excess Reactant.... That reactant which is not completely consumed in the rean.



Given $\begin{array}{|c|c|} \hline 10\text{g} & 10\text{g} \\ \hline \end{array}$

$$\begin{array}{cc} 10 - 1.25 & 10 - 10 = 0 \\ = 8.75\text{g} & \end{array}$$

For comp rean. of H_2 , O_2 required = $\frac{10 \times 32}{4} = 80\text{g}$ } X Not possible

For comp rean. of O_2 , H_2 required = $\frac{10 \times 4}{32} = 1.25\text{g}$ } ✓

L.R = O_2 , H_2 unreacted = $10 - 1.25\text{g} = 8.75\text{g}$

G.R = H_2 , H_2O produced = $\frac{10 \times 36}{32} = 11.25\text{g}$ } Ans

Shortcut for L.R....

→ In case of moles, Given moles of Reactant
Their stoichiometric coefficient
→ Reactant having lowest value will be L.R

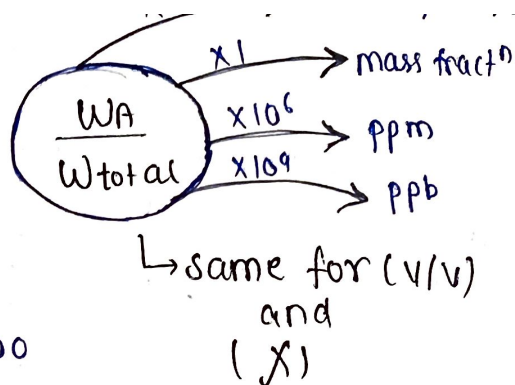
1. Percentage terms

* Mass percentage (w/w) = $\frac{W_A}{W_{total}} \times 100$

* Volume percentage (v/v) = $\frac{V_A}{V_{total}} \times 100$

* Mass by volume (w/v) = $\frac{W_A}{V_{total}} \times 100$

* Volume by mass (V/w) = $\frac{V_A}{W_{total}} \times 100$



Fraction terms

→ values lies b/w 0 to 1

a) Mass fraction

* M.F of A = $\frac{W_A}{W_A + W_B}$

* M.F of B = $\frac{W_B}{W_A + W_B}$

Mass fractⁿ of (A+B) = 1

$\frac{(M.F)_A}{(M.F)_B} = \frac{W_A}{W_B}$

b) Volume fraction

* V.F of A = $\frac{V_A}{V_A + V_B}$

* V.F of B = $\frac{V_B}{V_A + V_B}$

V.F of (A+B) = 1

$\frac{V_A}{V_A + V_B} + \frac{V_B}{V_A + V_B} = 1$

$\frac{(V.F)_A}{(V.F)_B} = \frac{V_A}{V_B}$

c) Mole fraction

* $X_A = \frac{n_A}{n_A + n_B}$

* $X_B = \frac{n_B}{n_A + n_B}$

$X_A + X_B = 1$

$\frac{X_A}{X_B} = \frac{n_A}{n_B}$

Parts term

1 million = 10^6
1 Billion = 10^9

a) Parts per million (ppm)

$(ppm)_A = \frac{W_A}{W_{total}} \times 10^6$
in terms of (w/w)

$(ppm)_A = \frac{V_A}{V_{total}} \times 10^6$
in terms of (v/v)

$(ppm)_A$ in terms of (w/v) = $\frac{W_A}{V_{total}} \times 10^6$

b) Parts per billion (ppb)

$(ppb)_A = \frac{W_A}{W_{total}} \times 10^9$
in term of (w/w)

$(ppb)_A = \frac{V_A}{V_{total}} \times 10^9$
in term of (v/v)

$(ppb)_A$ in term of (w/v) = $\frac{W_A}{V_{total}} \times 10^9$

Divide
 $\frac{\%}{\text{fract}^n} = 1$
 $\frac{ppm}{ppb} = \frac{10^6}{10^9}$

Strength... it represent mass of solute present in 1L/1000ml b/w

$$\text{Strength} = \frac{W_{\text{solute}}(\text{g})}{V_{\text{solution}}(\text{L})}$$

→ unit = g/L

→ Temp-dependent

Molarity (M)... it represent no. of moles of solute present in 1 L/1000 ml of soln.

$$n_{\text{solute}} = M \times V(\text{L})$$

$$\text{Molarity} = \frac{n_{\text{solute}}(\text{mol})}{V_{\text{soln}}(\text{L})}$$

→ unit = mol/L or Molar

→ temp. dependent

* Temp

If volume in ml:

$$M = \left(\frac{W_2}{M_2} \right) \times \frac{1000}{V(\text{ml})}$$

*

$$M = \frac{n_{\text{solute}} \times 1000}{V(\text{ml})}$$

$$n_{\text{solute}} = \frac{M \times V(\text{ml})}{1000}$$

W_2 = solute
 W_1 = solvent

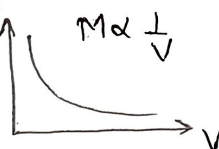
Dilution/concentration...

↓
Addition of solvent

↗ (n_{solute} = constant)
↘ Removal of solvent

$$M_1 V_1 = M_2 V_2$$

* * no. of millimoles of solute = $M \times V(\text{ml})$



Relⁿ b/w strength and Molarity...

$$\left[\text{Molarity} = \frac{\text{Strength}}{\text{Molar mass of solute}} \right]$$

$$M = \left(\frac{W_2}{M_2} \right) \times \frac{1000}{V(\text{ml})}$$

Relⁿ b/w Molarity and density of soln if % mass(w/w) of solute is given..

$$\text{Molarity} = \frac{10 \times \overbrace{(w/w)_{\text{solute}}}^x \times \overbrace{\text{density}}^d}{M_{\text{solute}}}$$

$$M = \frac{10 \times d}{M_{\text{solute}}}$$

Mixing... (n_{solute} = constⁿ)

$$M_1, V_1$$

$$M_2, V_2$$

$$M_1 V_1 + M_2 V_2 = M_{\text{mix}} V_{\text{mix}}$$

$$M_{\text{mix}} V_{\text{mix}}$$

→ alert in some questⁿ

$$M_{\text{mix}} = \frac{M_1 V_1 + M_2 V_2}{V_1 + V_2}$$

Molality...

• if mass in gram

$$\text{molality} = \frac{n_{\text{solute}}(\text{mol})}{W_{\text{solvent}}(\text{kg})}$$

→ unit = mol/kg or molar

→ temp. independent

*

$$\text{molality} = \frac{W_2}{M_2} \times \frac{1000}{W_1}$$

→ For numerical

15/10/2021
 Reln b/w Molality and Mole fraction....

$$\text{Molality} = \frac{n_{\text{solute}}}{n_{\text{solvent}}} \times \frac{1000}{M_{\text{solvent}}}$$

$$\text{Molality} = \frac{X_{\text{solute}} \times 1000}{X_{\text{solvent}} M_{\text{solvent}}}$$

Reln b/w Molality (m), Molarity (M) and Density of soln (d)...

$$\text{Molality (m)} = \frac{1000 \times \text{Molarity}}{1000 \times \text{density} - \text{Molarity} \times \text{Molar mass of solute}}$$

$$m = \frac{1000 M}{1000d - M \times M_{\text{solute}}} \rightarrow \text{molar mass}$$

Formality: Molarity for ionic solid

Normality:

$$\text{Dilution/concn} = \frac{N_1 V_1}{N_2 V_2}$$

$$\text{Normality (N)} = \frac{\text{gm. equivalent of solute}}{\text{Volume (l)}} \rightarrow \text{Temp. dependent}$$

Unit = eq./l or Normal

$$n_{\text{eq}} = n_{\text{soln}} \times n\text{-factor}$$

$$\text{no. of gmeq. of } x = \frac{\text{Given mass of } x (w)}{\text{Equivalent mass of } x (E)}$$

$$\text{Normality} = \text{Molarity} \times n\text{-factor}$$

$$\text{Equivalent mass} = \frac{\text{Molar mass}}{n\text{-factor}}$$

• When two Acid or Base are mixed:-

$$N_{\text{mix}} = \frac{N_1 V_1 + N_2 V_2}{V_{\text{mix}}}$$

• When acid and Base are mixed:-

$$N_{\text{mix}} = \frac{|N_A V_A - N_B V_B|}{V_{\text{mix}}}$$



Mercury → Solute

Thallium
or Na → solvent

Solubility...

Periodic Table

Elements

gas...
He
Ne
Ar
Kr
Xe
Rn

* Lanthanoid: $58-71 \rightarrow 4f \leftarrow$ Gp-3
* Actinoid: $90-103 \rightarrow 5f \leftarrow$ Gp-3

* Name of Element from 101 to 118

Mendeleev ko mila Nobel in Lawrence

Rutherford Dube sea me Bohr class Meit

Dar se Roe copper Niha fle to Mosco to Live Ten Ogan

2
8
8
18
18
32
32

For ns' [Rf Cu Ag Au Pt Cr Nb Mo Ru Rh] Pd, ns⁰ (n=5)

Rahgir Kyu Age Aau Pitayi Caroge Nababzaade maat Rukawat Rah Padegi

(n=4,5,6,7)

Rest all = s²

3-2 Sc²⁺ Ti²⁺ V²⁺ Cr²⁺ Mn²⁺ Fe²⁺ Co²⁺ Ni²⁺ Cu²⁺ Zn²⁺
3d¹ 3d² 3d³ 3d⁴ 3d⁵ 3d⁶ 3d⁷ 3d⁸ 3d⁹ 3d¹⁰

Man Tac Re $\rightarrow$ Mn, Tc, Re
Fen Ruya Osama $\rightarrow$ Fe, Ru, Os
Con Rhaga Iran me $\rightarrow$ Co, Rh, Ir

Gp-3 Sc Y La Ac - Sc, Y, La, Ac

Gp-7 Ni Padoge Pitoge - Ni Pd Pt

Gp-4 Tina Zara Haf Rafi - Ti, Zr, Hf, Rf

Gp-8 Cyu Aage Aau - Cu, Ag, Au

Gp-5 Von Nabab Tha Dabang - V, Nb, Ta, Db

Gp-9 Zindgi Cadbury Hg - Zn, Cd, Hg

Gp-6 Cuying Mohi Wite - Cr, Mo, W

Pseudo Inert Gas configuration - completely filled (n-1)d with filled ns

Zn, Cd, Hg

58 Ce: [Xe] 6s² 4f¹ 5d¹

64 Gd: [Xe] 6s² 4f⁷ 5d¹

71 Lu: [Xe] 6s² 4f¹⁴ 5d¹

59 = 4f¹
62 = 4f⁷
65 = 4f⁷
70 = 4f¹⁴
2:3:2
1 7 14

Z_{eff} $\propto$ no. of proton
no. of electron

Penetration effect: s > p > d > f

closer to nucleus

Shielding effect: s > p > d > f

(s) $\propto$ (Z_{eff})²

Z_{eff}: Al < Ga

Z_{eff}: Cd < Hg

D.E.: Cd < Hg

Atomic Radius

Covalent Radius

Metallic Radius

Vanderwaal Radius

(Non-metal)

(Metal)

(molecule & inert gas)

r_c < r_m < r_v

Ni >> Li > Be > B > C > N

* Same Period - L $\rightarrow$ R; size $\downarrow$

* Diff. Period - Period $\uparrow$ size $\uparrow$

Ionic Radius $\propto$ -ve charge / +ve charge

Ionic Radius $\propto$ 1 / Z_{eff}

* O²⁻ < N³⁻ < S²⁺ < P³⁻
n=2 n=3

** B < Ga < Al < In < Tl; size

** B³⁺ < Al³⁺ < Ga³⁺ < In³⁺ < Tl³⁺; Ionic size

4f series: (Eu) > Ce > Pr > Nd > Pm $\rightarrow$ Lu

Ionic Radius - same

Lanthanide contraction

Ionic radius: Ce³⁺ > Pr³⁺ > Nd³⁺ > Pm³⁺

Actinide contraction - size - $T \rightarrow B > \dots > L \rightarrow R$
 $D.R - T \rightarrow B > \dots > L \rightarrow R$ shielding; $4f > 5f$

Atomic size of D-block element - 3d series < 4d series < 5d series

- Element in liq. state at room temp: ^{Metal} Hg ^{Non-metal} Br₂
- Metals in liq. state at $T < 40^\circ C$: Rb, Hg, Ga, Pb, Bi, Sn $\rightarrow$ Rubimar gyi Fir se ^{High Temp.}
- Rare earth metal: f-block
- Trans Uranium element: $Z > 92$
- Radioactive element: 84 to 118 + ^{Lanthanide} Pm + Tc; All 14 elements are radioactive in Actinoids
- Element which exhibit both vertical and horizontal similarities; Transition element

Ionisation Energy $\propto Z_{eff} \propto \frac{+ve}{-ve}$

$\rightarrow DE_1 < DE_2 < DE_3 \dots$
 $\rightarrow$ Top $\rightarrow$ bottom = size $\uparrow$; $DE \downarrow$
 $\rightarrow$ left $\rightarrow$ right = $DE \uparrow$; metallic character $\downarrow$

$DE_1 : N > O$
 $DE_2 : N < O$
 $DE_3 : As > N < O >$

$DE = Li < B < Be < C < O < N < F < Ne$ * $IE \uparrow$
 Highest DE : He
 Lowest DE : Cs

For 3d series: $Zn > Fe > Co > Cu > Ni > Mn > Ti > Cr > V > Sc$

Zaan fir ko baat kyu nhi Maanti (Zn is very good scam)

No. of valence e^- = No. of n th ionization energy just before sudden jump

Metallic Character/Reactivity: $T \rightarrow B$ (Reactivity $\uparrow$)

$L \rightarrow R$ (Reactivity $\downarrow$)

Electron Gain Enthalpy...

$\uparrow \uparrow$ FOA; Release of Energy; stability $\uparrow$; Energy $\downarrow$ ($\Delta H_{eg} = -ve$)

$\uparrow \uparrow$ FOR; Absorption of Energy; stability $\downarrow$; Energy $\uparrow$ ($\Delta H_{eg} = +ve$)

Cl = Most electronegative

Ne = most electropositive

Electron Affinity...

$\rightarrow Z_{eff} L \rightarrow R$; $Z \uparrow$, $Z_{eff} \uparrow$, $EA \uparrow$, Electronegativity $\uparrow$

$\rightarrow T \rightarrow B$; Z_{eff} constn, size $\uparrow$, $EA \downarrow$

$\rightarrow$ closeness to nucleus $\uparrow$, $EA \uparrow$

EA : $grp-1 > grp-13$

B C N O F (2nd period)
 $\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$
 Al Si P S Cl (3rd period)

* For Inert gas and Stable electronic configuration both have no affinity for e^-

$2p-2p > 2p-3p > 3p-3p$

* $-ve \Delta H_{eg}$ for Alkali metals and Hydrogen; EA : $H > Li > Na > K > Rb > Cs$

* $EA = Cl > F > Br > I > At$ e^- element

EA : $S > Se > Te > Po > O$

* $grp-16 < grp-17$
 $S < At$

* $EA = Ne > Ar > Kr > Xe > Rn > He$ e^+ element

Reln b/w DE and E_{eg}... Most imp...

$$|D.E. of Na| = |\Delta E_{eg} of Na^+|$$

$$\Delta E_{eg} = F < Cl$$

$$D.E. = F^- < Cl^-$$

$$D.E. = F > Cl$$

$$\Delta E_{eg} = F^+ > Cl^+$$

$$E.A. = F < F^+$$

$$Cl < Cl^+$$

$$D.E. = Cl > Cl^-$$

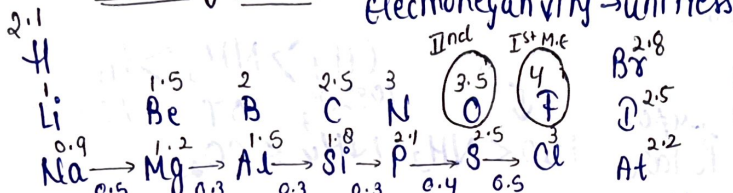
$$F > F^-$$

$$E.A.: F^+ > Cl^+ > Cl > F$$

$$D.E.: F^- < Cl^- < Cl < F$$

Pauling Scale...

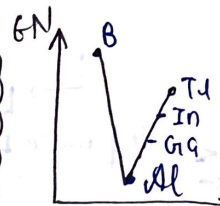
Electronegativity → unitless



$$H, P = 2.1$$

$$C, I, S = 2.5$$

$$N, Cl = 3$$



$$EN: B > N > F > O > Cl > S > C > Si > Ge > Sn > Pb$$

$$C > Pb > Si > Ge > Sn$$

Mulliken Scale...

$$X_M = 2.8 X_P$$

↓
E.N. of Mulliken E.N. of Pauling

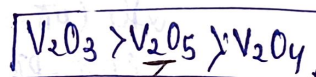
* $\Delta E.N \uparrow$; charge $\uparrow$;
Ionic character $\uparrow$; Covalent character $\downarrow$

Diagonal Relationship...



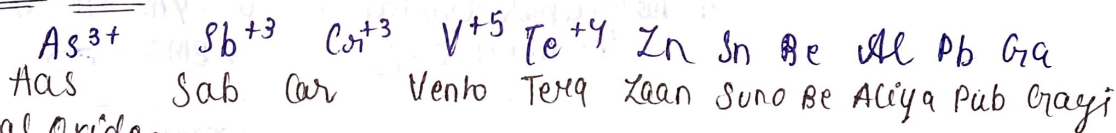
* Draxises bcz of similar size of atom and ion.

Oxides →
→ 3. Acidic (+5, +6, +7)
→ 1. Basic (+1, +2, +3)
→ 2. Amphoteric (+4)



Amphoteric but Acidic predominately

Amphoteric Oxide... # V_2O_4 (less basic)



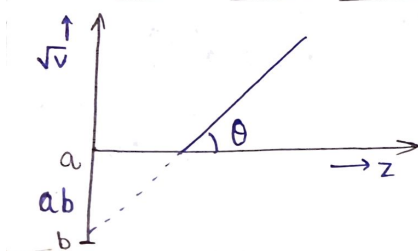
Neutral Oxide...
NO / N₂O / CO

Basic Oxide = Metals

Acidic Oxide = Non-metal

Left → Right: Non-metallic character $\uparrow$
~~Acidic character $\uparrow$~~

Modern Periodic Table...



$$\tan \theta = m = \text{slope} = a$$

$$\# \sqrt{v} \propto z$$

$$\sqrt{v} = a(z-b)$$

$$\sqrt{v} = az - ab$$

→ Moseley's eqn

* 94 naturally occurring elements

* Pitch blende → An ore of Uranium

* Small cation: H⁺

Small anion: F⁻

* $F^- < Cl^- < Br^- < I^-$: size
* Eka Al = Ga ; Eka B = Sc ; Eka Si = Ge

Unit and Dimension

Fundamental Units....

Length, Mass, Time, Temperature, Elec. Current, Luminous intensity, Amount of Substance.

Supplementary Units....

Plane Angle, Solid Angle

Both are dimensionless but have unit

Significant figures...

- * All non-zero digits are significant
- * Zero b/n two non-zero digits is significant
- * Leading zero or zeros b/w placed to left are never significant.
- * Trailing zeroes or zero placed to the right of the number after decimal are significant.
- * In exponential expression the numerical portion given the number of S.F.

Dimensional Analysis...

- * Dimension of physical quantity are power to which units of base quantity are raised.

$$\text{eg: } [M]^a [L]^b [T]^c [A]^d [K]^e$$

- 1) Checking the correctness of various formulae -

$$\text{eg: } Z = A + B$$

$$Z = [A] = [B]$$

- 2) Conversion of one system of unit to other

$$n_1 U_1 = n_2 U_2$$

$$\text{eg: } n_1 [M_1^a L_1^b T_1^c] = n_2 [M_2^a L_2^b T_2^c]$$

$$n_1 = n_2 \left[\frac{M_2}{M_1} \right]^a \left[\frac{L_2}{L_1} \right]^b \left[\frac{T_2}{T_1} \right]^c$$

- Mass $\longrightarrow M$
- Length $\longrightarrow L$
- Time $\longrightarrow T$
- Velocity $\longrightarrow LT^{-1}$
- Acceleration $\longrightarrow LT^{-2}$
- Force $\longrightarrow MLT^{-2}$
- Energy $\longrightarrow ML^2T^{-2}$
- Power $\longrightarrow ML^2T^{-3}$
- Force gradient $\longrightarrow MT^{-2}$

- 3) Formula of force to find dimⁿ of different PQ.

$$\longrightarrow F = G \frac{m_1 m_2}{r^2}, F = \frac{q_1 q_2}{4\pi \epsilon_0 r^2}$$

$$\longrightarrow F = Kx, F = \frac{\mu_0 I_1 I_2 d}{4\pi r}$$

$$\longrightarrow F = qE$$

$$\longrightarrow F = qVB \longrightarrow F = SI$$

$$\longrightarrow F = 6\pi\eta r v$$

Dimensionless Quantities.....

Strain, Refractive index, Relative density, Exponential function
Trigonometry function, Relative permittivity, Pure number
Efficiency, Length gradient, coef. of friction.

* Formula of Energy to find dim of diff. physical Qty

$$\begin{aligned} \rightarrow E &= hf & \rightarrow H &= I^2 R t & \rightarrow \frac{H}{t} &= \frac{K A \Delta T}{l} \\ \rightarrow E &= \frac{3}{2} k_B T & \rightarrow E &= \frac{1}{2} L I^2 & \rightarrow P V &= n R T & \rightarrow \text{Stress} &= \gamma \times \text{Strain} \\ \rightarrow E &= \frac{Q^2}{2C} & \rightarrow Q &= m s \Delta T & \rightarrow Q &= m L \end{aligned}$$

<u>Dimension</u>	May have dim ⁿ or may be dim ⁿ less	Must be dimensionless	c) A P.Q have dimension	d) A P.Q does not have dimension
<u>Unit</u>	a) A physical quantity have unit	b) A physical quantity does not have unit	Must have unit	May or may not have unit

<u>Physically correctness</u>	Must be physically wrong	May or may not be physically correct	c) Equation is physically wrong	d) Equation is physically correct
<u>Dimensional correctness</u>	a) Equation is dimensional wrong	b) Equation is dimensional correct	May or may not be dimensionally correct	Must be dimensionally correct

$$S_n = u + \frac{a}{2} (2n - 1)$$

↳ dimensionally correct bcz it is displacement in sec

Instruments.....

Least Count :- mm scale

↓
1mm

Vernier scale

↓
0.1mm

Screw Gauge

↓
0.01mm

Vernier callipers -

$$L.C = 1 \text{ MSD} - 1 \text{ VSD}$$

↓
Main scale Division Vernier scale Division

LSR = Linear scale Reading
CSR = Circular scale Reading

Screw Gauge - Pitch = $\frac{\text{MSR}}{\text{no. of rotation}}$; L.C = $\frac{\text{Pitch}}{\text{Total no. of division on circular scale}}$

$$\text{Total Reading} = 1 \text{ LSR} + \text{CSR} \times \text{L.C}$$

Accuracy - It is the measure of how close the measured value is to the true value. Closeness of measured & true value.

Precision - It tells us to what resolution or limit the quantity is measured.

Error in Measurement.

Absolute Error - $\Delta a = |a_i - a_{\text{mean}}|$

- Always positive $a_{\text{mean}} = \frac{a_1 + a_2 + \dots + a_n}{n}$
- Unit and dimⁿ same as P.Q.
- Least count error can be taken as absolute error.
- It cannot tell about accuracy of measurement

Relative Error - $\frac{\Delta a_{\text{mean}}}{a_{\text{mean}}}$

- Unit and dim^{less}
- It tells about accuracy of measurement
- Random error can be decreased by increasing no. of observations

$$\Delta a_{\text{mean}} = \frac{\Delta a_1 + \Delta a_2 + \dots + \Delta a_n}{n}$$

$$\text{Percentage Error} = \frac{\Delta a_{\text{mean}}}{a_{\text{mean}}} \times 100$$

General Rules.....

$$Z = \frac{A^p B^q}{C^r}$$

Then max. fracⁿ relative error in Z will be-

$$\frac{\Delta Z}{Z} = p \frac{\Delta A}{A} + q \frac{\Delta B}{B} + r \frac{\Delta C}{C}$$

Important -

- Addition/Subtraction me pahle absolute error nikalenge phir relative.
- Power/multiplication/division me pahle relative error nikalenge phir absolute.

Motion in a plane.....

SHORT NOTES

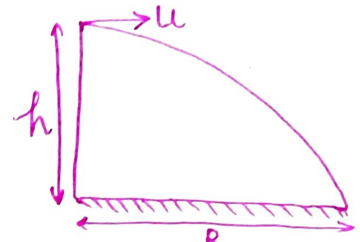
Projection Motion....

Horizontal Projection from a height h

* Time of flight $\Rightarrow T = \sqrt{\frac{2h}{g}}$

* Horizontal Range $\Rightarrow R = UT = u\sqrt{\frac{2h}{g}}$

* Angle of velocity at any instant with horizontal: $\theta = \tan^{-1}\left(\frac{gt}{u}\right)$



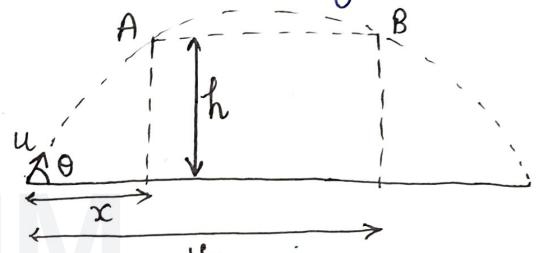
Projectile Motion....

A body crosses two points at same height in time t_1 and t_2 , the points are at distance x and y from starting point then

$\rightarrow x + y = R$

$\rightarrow t_1 + t_2 = T$

$\rightarrow h = \frac{1}{2}gt_1t_2$



Velocity of particle at Time t during projectile....

$$\vec{V} = V_x \hat{i} + V_y \hat{j} = U_x \hat{i} + (U_y - gt) \hat{j} = U \cos \theta \hat{i} + (U \sin \theta - gt) \hat{j}$$

* If angle of velocity $\vec{V}$ from horizontal is α , then

$$\tan \alpha = \frac{V_y}{V_x} = \frac{U_y - gt}{U_x} = \frac{U \sin \theta - gt}{U \cos \theta} = \tan \theta - \frac{gt}{U \cos \theta}$$

At highest point: $V_y = 0, V_x = U \cos \theta$

Time of flight: $T = \frac{2U_y}{g} = \frac{2U \sin \theta}{g}$

Horizontal Range....

$$R = (U \cos \theta) T = \frac{2U^2 \sin \theta \cos \theta}{g} = \frac{U^2 \sin 2\theta}{g} = \frac{2U_x U_y}{g}$$

* It is same for θ and $(90^\circ - \theta)$ and max^m for $\theta = 45^\circ$

Maximum Height: $H = \frac{U_y^2}{2g} = \frac{U^2 \sin^2 \theta}{2g} = \frac{1}{8} g T^2, \frac{H}{R} = \frac{1}{4} \tan \theta$

Equation of trajectory -

$$y = x \tan \theta - \frac{g x^2}{2 u^2 \cos^2 \theta} = x \tan \theta \left(1 - \frac{x}{R}\right)$$

Relative Motion....

→ Relative velocity of Rain w.r.t the Moving Man.

→ The relative velocity of rain w.r.t man

$\vec{V}_{rm} = \vec{V}_r - \vec{V}_m$ will be represented

by diagonal $\overline{OD}$

$$\therefore V_{rm} = \sqrt{V_r^2 + V_m^2 + 2 V_r V_m \cos 90^\circ}$$

$$V_{rm} = \sqrt{V_r^2 + V_m^2}$$

→ If θ is the angle which $\vec{V}_{rm}$ makes with the vertical direction then -

$$\tan \theta = \frac{BD}{OB} = \frac{V_m}{V_r} \Rightarrow \theta = \tan^{-1} \left(\frac{V_m}{V_r} \right)$$

Swimming into the Water....

* If the swimming is in the direction of flow of water or along the downstream then

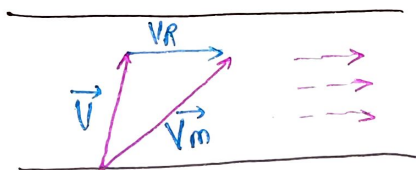
$$\vec{V} \quad \vec{V}_R \quad V_m = V + V_R$$

* If the swimming is in the direction opposite to the flow of water or along the upstream then

$$\vec{V} \quad \vec{V}_R \quad V_m = V - V_R$$

* If man is crossing the river as shown $\vec{V}$ and $\vec{V}_R$ are non-collinear then use the vector algebra

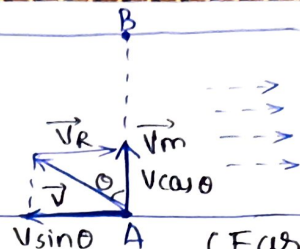
$$\vec{V}_m = \vec{V} + \vec{V}_R \quad (\text{assuming } V > V_R)$$



For shortest path -

* To reach at B,

$$V \sin \theta = V_R \Rightarrow \sin \theta = \frac{V_R}{V} \frac{d}{d}$$

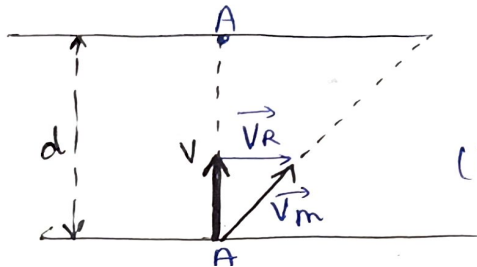


(For min displacement)

* Time of crossing, $t = \frac{d \cos \theta}{V}$

For minimum time -

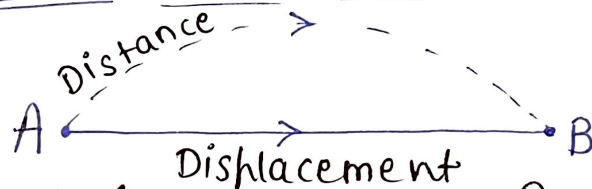
$$\text{Drift} = t_{\min} V_R = \frac{d}{V} (V_R)$$



(For min. time)

Motion in a straight line...

Distance v/s Displacement...



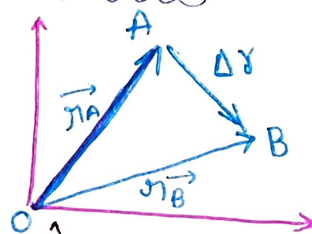
Displacement is change of Position Vector...

From ΔOAB : $\Delta \vec{r} = \vec{r}_B - \vec{r}_A$

$$\vec{r}_B = x_2 \hat{i} + y_2 \hat{j} + z_2 \hat{k}$$

$$\vec{r}_A = (x_2 - x_1) \hat{i} + (y_2 - y_1) \hat{j} + (z_2 - z_1) \hat{k}$$

$$\Delta \vec{r} = (x_2 - x_1) \hat{i} + (y_2 - y_1) \hat{j} + (z_2 - z_1) \hat{k}$$



Average velocity = $\frac{\text{Displacement}}{\text{Time Interval}} \Rightarrow \vec{V} = \frac{\Delta \vec{r}}{\Delta t}$

Average speed = $\frac{\text{Distance travelled}}{\text{Time Interval}}$

For Uniform Motion...

Average speed = |average velocity| = |instantaneous velocity|

$$\text{velocity} = \frac{d\vec{r}}{dt} = \frac{d}{dt} (x\hat{i} + y\hat{j} + z\hat{k})$$

$$\vec{v} = \frac{dx}{dt} \hat{i} + \frac{dy}{dt} \hat{j} + \frac{dz}{dt} \hat{k}$$

$$\boxed{\vec{v} = v_x \hat{i} + v_y \hat{j} + v_z \hat{k}}$$

Average accⁿ = $\frac{\text{change in velocity}}{\text{Time}} = \frac{\Delta \vec{v}}{\Delta t}$

Acceleration = $a_x \hat{i} + a_y \hat{j} + a_z \hat{k}$

Important points About 1D Motion....

- Distance $\geq$ |Displacement|
- Average speed $\geq$ |Average velocity|
- If distance > |displacement| this implies atleast at one point in path, velocity is zero.



Motion with Constant Acceleration....

* $V = U + at$

* $S = \left(\frac{U+V}{2}\right)t = ut + \frac{1}{2}at^2 = vt - \frac{1}{2}at^2$

* $V^2 = U^2 + 2as$ * $S_n = U + \frac{a}{2}(2n-1)$

Motion under gravity....

- If an object is falling freely ($u=0$) under gravity, then equation of motion becomes -

i) $V = u + gt$ ii) $h = ut + \frac{1}{2}gt^2$

iii) $V^2 = u^2 + 2gh$

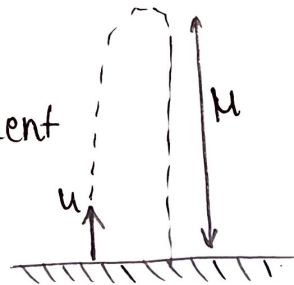
Note: If an object is thrown upward then g is replaced by $(-g)$ in above three equation-

- If a body is thrown vertically up with a velocity u in the uniform gravitational field then

i) Max^m height attained $H = \frac{u^2}{2g}$

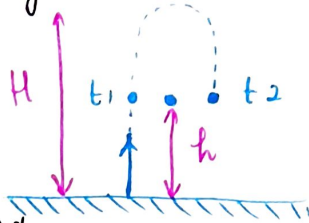
ii) Time of ascent = time of descent
= $\frac{u}{g}$

iii) Total time of flight = $\frac{2u}{g}$



- iv) Velocity of fall at the point of projection = u (downward)
 v) At any point on its path the body will have same speed for upward journey and downward journey.

max. height $\boxed{H = \frac{1}{8} g (t_1 + t_2)^2}$

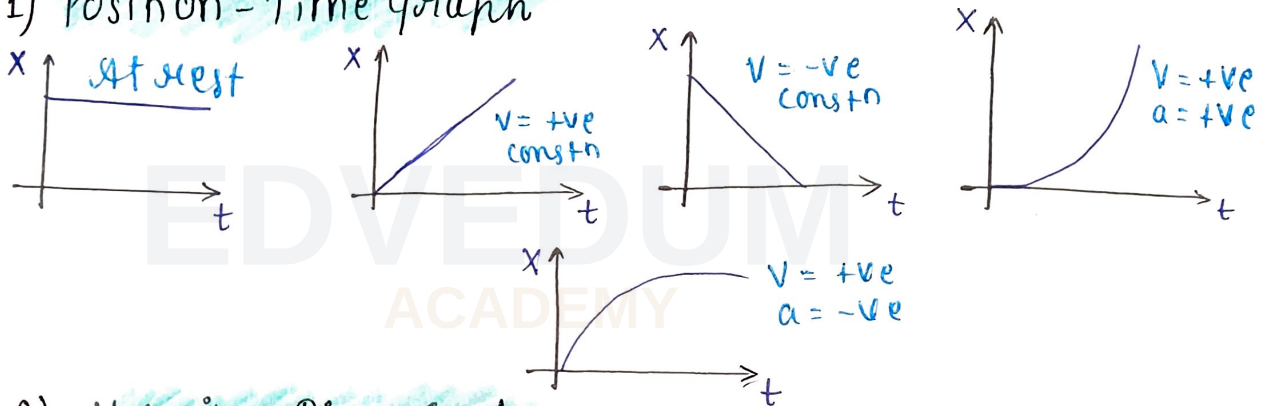


* A body is thrown upward, downward and horizontally with same speed takes time t_1 , t_2 & t_3 respectively to reach the ground then $t_3 = \sqrt{t_1 t_2}$ and height from where the particle was throw is

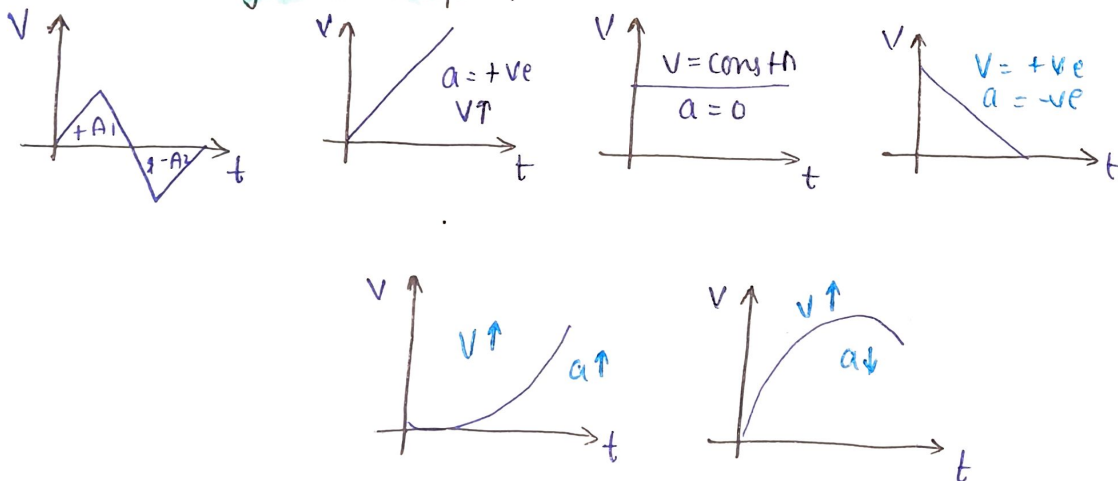
$$\boxed{H = \frac{1}{2} g t_1 t_2}$$

Graph...

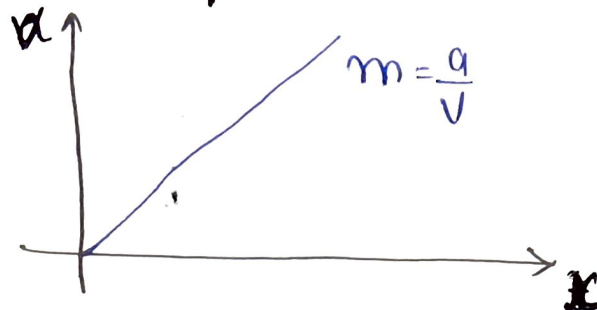
1) Position - Time Graph



2) Velocity - Time Graph



3) Velocity - Position graph



4) Acceleration-time graph

